

Assignment 2

Deadline: 11 November, 2025

There are three questions provided below. Question 2 and 3 are optional for submission. You may consider question 2 and 3 as practice questions.

Question 1:

Given the following grammar:

$S \Rightarrow A;$

$A \Rightarrow BC$

$A \Rightarrow D$

$B \Rightarrow \text{int}$

$B \Rightarrow \text{float}$

$C \Rightarrow \text{id}$

$D \Rightarrow ()$

- Construct the LR(1) Automaton for the given grammar.
- Build the CLR(1) Parsing Table based on the LR(1) items.
- Simulate the parsing process using a stack for the input string:
int id;

Question 2:

[Optional Submission, You may consider this as practice]

Given the following grammar:

$S \Rightarrow A \text{ id } B;$

$A \Rightarrow \text{int} \mid \text{float}$

$B \Rightarrow BC \mid \epsilon$

$C \Rightarrow [\text{num}]$

- Construct the LR(o) Automaton for the given grammar.
- Build the SLR(1) Parsing Table based on the LR(o) items.
- Simulate the Parsing Process using a stack for the input string:
float id [num] [num];

Question 3:

[Optional Submission, You may consider this as practice]

Given the following grammar:

$V \Rightarrow PQ$

$P \Rightarrow \text{int}$

$P \Rightarrow \text{float}$

$Q \rightarrow \epsilon$

$Q \rightarrow [\text{size}] Q$

- d. Construct the LR(o) Automaton for the given grammar.
- e. Build the SLR(1) Parsing Table based on the LR(o) items.
- f. Simulate the Parsing Process using a stack for the input string:
float [size] [size] [size];

[Note: You can verify your solution on BUX, where the solution for this grammar is provided under [Lecture 7](#)]